

MAHESHA S



📍 Mysuru, India 570022

📞 +91 98802 77334

✉ mahesha.shivamallappa@gmail.com

✂: Compilers • Computer Architecture • AI Accelerator • GPU • LLVM • MLIR

This detailed resume is organized so you can explore Mahesha's 20+ year compiler engineering career, along with his perspective on leadership, at whatever depth is useful to you — from a quick career summary to a fully detailed technical view. Click a heading below to jump straight to the section you want to explore.

1	<u>One-Page Resume</u>	A concise summary of experience, roles, and education at a glance.
2	<u>Career Timeline</u>	A transparent record of employment dates, transitions, and career breaks — for anyone verifying continuity.
3	<u>Engineering Portfolio</u>	A detailed, problem-by-problem account of the technical work behind each role — for anyone looking for technical depth.
4	<u>LinkedIn Profile</u>	A view into professional activity and network — for anyone who wants the broader picture.
5	<u>Technical Papers</u>	Published technical papers and write-ups — for anyone interested in the technical writing behind the work.
6	<u>My Thoughts About Leadership</u>	A personal essay on what genuine leadership demands — for anyone curious about the philosophy behind how I lead and mentor.

One-Page Resume

A concise, one-page overview of professional experience, roles, and education.

ABOUT ME

Compiler engineer with 20+ years of experience building production compilers across the stack—from Clang and MLIR through LLVM—for CPU, GPU, and AI accelerator architectures. I love tackling hard problems, figuring out how things really work, and building solutions from scratch. I also enjoy helping the engineers around me grow into strong technical leaders.

PROFESSIONAL SUMMARY

INTEL Bangalore
Compiler Engineer • Jun 2022 – Aug 2026

MLIR/LLVM based AI/GPU compiler.

AMD Bangalore
Compiler Engineer • Feb 2019 – Jun 2022

LLVM based AMD GPU compiler.

QUBOLE Bangalore
Software Engineer • May 2018 – Jan 2019

Distributed deep learning infrastructure on Qubole data lake platform.

JIVOX Bangalore
Software Engineer • Jul 2016 – Feb 2018

Recommender engine for Jivox Ad personalization product.

SAMSUNG Bangalore
Compiler Engineer • Apr 2013 – Apr 2015

Randomizer for automated testing of OpenCL compiler.

AMD Bangalore
Compiler Engineer • Jul 2012 – Mar 2013

LLVM based C++ compiler and OpenMP support in Clang front-end.

SIEMENS Bangalore
Research Engineer • Jul 2008 – Jun 2012

Heterogeneous computing research and compiler for Siemens PLCs.

HP Bangalore
Compiler Engineer • Sep 2005 – Jun 2008

Proprietary C++ compiler for HP PA-RISC architecture.

DACS SOFTWARE Bangalore
Software Engineer • Jun 2004 – May 2005

Assembly-to-assembly code translator tool for Ceva Inc DSP processors.

EDUCATION

MASTER'S University of Mysore
Jul 2000 – Jun 2003

Master's in Computer Science

BACHELOR'S University of Mysore
Jul 1996 – Jun 2000

Bachelor's in Mathematics

Career Continuity & Timeline

*A transparent record of professional continuity, career transitions, and engineering growth across two decades.
(As of 15 Aug 2026)*

Company	Tenure (From–To)	Career Transition	Duration (y-m-d)	Career Break (y-m-d)
DACS	2004/06/14 2005/05/23	Pursued long-term compiler engineering opportunities.	00y-11m-10d	00y-03m-22d
HP	2005/09/19 2008/06/17	Accepted programming languages and compiler research opportunity.	02y-09m-00d	00y-00m-19d
SIEMENS	2008/07/07 2012/06/22	Accepted broader compiler engineering responsibilities.	03y-11m-16d	00y-00m-09d
AMD	2012/07/02 2013/03/28	Role concluded following corporate restructuring.	00y-08m-27d	00y-00m-17d
SAMSUNG	2013/04/15 2015/04/15	Planned career break for family responsibilities and long-term career reflection.	02y-00m-00d	01y-02m-18d
Phase 1 Completed — First decade of industry experience				
JIVOX	2016/07/04 2018/02/06	Pursued opportunities in large-scale machine learning systems.	01y-07m-03d	00y-03m-14d
QUBOLE	2018/05/21 2019/01/25	Returned to compiler engineering aligned with long-term career goals.	00y-08m-05d	00y-00m-09d
AMD	2019/02/04 2022/06/17	Accepted an AI compiler engineering opportunity.	03y-04m-14d	00y-00m-02d
INTEL	2022/06/20 2026/08/15	Role concluded following corporate restructuring.	04y-01m-27d	
Phase 2 Completed — Twenty years of industry experience				
Total			20y-02m-12d	01y-11m-20d

Engineering Portfolio

A reverse-chronological record of the engineering problems, technical challenges, and corresponding solutions across a 20+ year compiler engineering career, from VLIW assembly-to-assembly code translation at DACS Software to Intel's AI/GPU compiler stack.

Intel, Bangalore		
Compiler Engineer • Jun 2022 – Aug 2026		
1	PROBLEM	Intel's AI accelerator originally represented tensor dimensions with a 5-d vector register ("dim5"), each dimension a 32-bit value matching the accelerator's 32-bit general-purpose registers (GPRs). As LLM weight counts started to grow, Intel widened each tensor dimension to 44 bits while keeping GPRs at 32 bits — a width with no natural, well-supported type in the LLVM/Clang type system. The compiler needed to expose this new register width to developers without (a) introducing a 44-bit language-level type, (b) disrupting the existing dim5 programming model, or (c) requiring hardware-specific reasoning at the language level.
	SOLUTION	- Exposed the hardware's 44-bit tensor register at the language level as a standard 64-bit type — since 64 bits map naturally onto a pair of 32-bit GPRs — so the existing dim5 type transparently became a 5-dimension vector of 64-bit entries, and both the front-end and middle-end could continue operating on a single, standard-width type. - During instruction selection, lowered 64-bit tensor operations to the hardware's actual 44-bit width using two-step semantics: (a) splitting each 64-bit value into its low 32-bit and high 12-bit halves and mapping them onto an even/odd pair of 32-bit GPRs, and (b) using the hardware's native low/high copy and data-movement semantics. - Implemented targeted post-instruction-selection optimization passes to eliminate redundant register moves introduced by the low/high emulation, keeping the generated code efficient despite the additional emulation overhead.
2	PROBLEM	The 44-bit, 5-d tensor register (dim5) support above shipped as production code, but like any new feature, it surfaced bugs and rough edges as customer adoption grew that needed to be resolved.
	SOLUTION	- Stabilized the compiler infrastructure by resolving customer-reported issues, strengthening automated test coverage, and improving code maintainability through systematic refactoring. - Further optimized the 44-bit tensor code-generation passes for improved efficiency.
3	PROBLEM	The team was reassigned from AI accelerator compiler work to GPU compiler development. Its engineers needed to quickly ramp up on GPU architecture and compiler requirements to become productive contributors.
	SOLUTION	- Ran knowledge-sharing sessions to build the team's working understanding of core GPU architecture and key hardware-level details (e.g., warps, occupancy, memory hierarchy). - Prototyped a divergence analysis pass for the GPU compiler and walked the team through the underlying concepts and implementation details. - Researched IEEE 754 and emerging low-precision floating-point formats and presented their practical compiler implications to the team.
4	PROBLEM	Automatically generating efficient GPU kernels from PyTorch computation graphs depends on robust graph fusion. The newly introduced MLIR-based Fuser needed to be stabilized before it could serve as a reliable foundation for further MLIR-based fusion development.

	SOLUTION	• Moved into the MLIR ecosystem and stabilized the MLIR Fuser by diagnosing and resolving critical functional bugs across the compilation pipeline. • Improved compiler robustness through targeted refactoring and issue resolution across multiple phases of the MLIR-based graph fusion pipeline.
5	PROBLEM	Low-precision numerical formats (e.g., FP8, FP4) have become central to AI training and inference due to performance and memory efficiency requirements, so the MLIR Fuser needed end-to-end support for these formats — including scale computation, quantization, and scaled matrix-multiply-accumulate (scaled MMA) lowering — across the full MLIR compiler stack.
	SOLUTION	• Designed and implemented end-to-end compiler support for low-precision floating-point types throughout the MLIR Fuser pipeline. • Resolved technical challenges arising from sub-byte data representations, evolving GPU hardware capabilities, inconsistent type representations across AI frameworks, MLIR, LLVM, and GPU architectures, and the fast-evolving nature of low-precision support across the AI stack. • Delivered scale-computation, quantization, and scaled-MMA lowering implementations designed to extend cleanly as the low-precision AI hardware ecosystem continued to evolve.

AMD, Bangalore

Compiler Engineer • Feb 2019 – Jun 2022

1	PROBLEM	AMD was investing in a robust open-source GPU programming model to grow the ROCm ecosystem as an open alternative to CUDA. The Clang-based HIP-Clang compiler was central to that strategy and needed rapid stabilization to support growing customer adoption.
	SOLUTION	• Stabilized the HIP-Clang compiler by resolving customer-reported issues and implementing requested language features. • Worked with the internal PyTorch team to establish HIP-Clang as the default compiler and toolchain for PyTorch's AMD GPU (HIP) backend. • Fixed HIP runtime issues that were blocking HIP-Clang's productization.
2	PROBLEM	AMD needed to support Cooperative Groups, a GPU programming model that lets kernels form and synchronize thread groups beyond the traditional thread-block (work-group) granularity. These groups can span multiple levels of the execution hierarchy — warp-level sub-groups, thread-blocks, and the grid. This required the device library to expose efficient, low-level abstractions for these hierarchy-spanning group handles and their member functions.
	SOLUTION	• Worked with the GPU hardware team to design and implement device-library support for Cooperative Groups, spanning the full execution hierarchy — warp-level sub-groups, thread-blocks, and the grid. • Implemented core handles and member functions — including <code>thread_rank()</code>, <code>num_threads()</code>, and <code>thread_index()</code> — as efficient hardware abstractions usable consistently across each level of that hierarchy.
3	PROBLEM	On AMD GPUs, shared memory (LDS) is a kernel-scoped resource, so a non-kernel (device) function can't independently declare its own shared-memory variables — the compiler has to arrange for that memory to exist by the time the function runs. AMD's existing module-lds-lowering pass did this by consolidating every shared-memory variable reachable from a kernel, across all of its call paths, into a single structure allocated once per kernel. This was functionally correct, but it sized the structure for the worst case: large shared-memory arrays used only on rarely-taken paths still consumed LDS space on every kernel launch, creating a direct trade-off between functionality and memory efficiency.

	SOLUTION	- Designed and implemented an LLVM/AMDGPU compiler pass that replaced non-kernel-function uses of LDS (shared-memory) globals with pointer indirection — each pointer computed once inside the owning kernel and reconstructed inside the non-kernel function — so the module-lowering pass no longer needed to pack the full LDS object into every kernel that could reach it, significantly reducing per-kernel LDS memory footprint. - The patch was accepted and merged into upstream LLVM, but was later disabled by default after it caused regressions in OpenMP workloads.
	PROBLEM	AMD's strategic collaboration with CERN required successfully migrating CERN's GPU applications to the ROCm ecosystem — with both functional correctness and competitive performance.
4	SOLUTION	- Worked across multiple ROCm teams to get CERN's GPU applications compiling and running correctly on the ROCm ecosystem. - Improved application performance through coordinated optimization work spanning both the AMD GPU compiler backend and CERN's application source code.
	PROBLEM	With ROCm adoption growing, application performance became a key customer expectation, which required ongoing compiler improvements to resolve workload-specific performance bottlenecks.
5	SOLUTION	- Investigated and resolved customer-reported performance issues through targeted compiler optimizations. - Improved generated code quality by refining low-level optimization passes — including instruction scheduling and register allocation — to meet customer-expected performance.

Qubole, Bangalore

Software Engineer • May 2018 – Jan 2019

	PROBLEM	With deep learning frameworks rapidly gaining adoption, Qubole saw an opportunity to extend its cloud-native data-lake platform beyond traditional big-data workloads by offering distributed deep learning as a managed service. This required evaluating how modern deep learning frameworks could integrate with Qubole's existing distributed infrastructure.
1	SOLUTION	- Worked with multiple engineering teams to evaluate how distributed deep learning frameworks could be integrated with Qubole's existing data-lake infrastructure. - Built and validated a rapid prototype demonstrating the platform's ability to support deep learning as a managed cloud service, informing architecture discussions on production feasibility.

Jivox, Bangalore

Software Engineer • Jul 2016 – Feb 2018

1	PROBLEM	Jivox saw an opportunity to strengthen its ad-personalization platform with machine learning-based recommendations as ML gained prominence in digital advertising. This required integrating recommendation capabilities into the existing personalization engine without disrupting production stability.
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	SOLUTION	- Designed and integrated a recommender-system-based ad-personalization engine into the existing platform. - Iteratively refined the recommendation logic based on customer feedback and production performance data. - Stabilized the production system through systematic issue investigation and resolution.
	PROBLEM	As user and product data volume grew rapidly, scaling the ad-personalization platform required sharding data efficiently across multiple database servers.
2	SOLUTION	- Designed and implemented a sharding algorithm to distribute user and product data across multiple database servers, and integrated it into the existing ad-personalization platform. - Improved the sharding infrastructure over time through production issue resolution and logic refinement.

Samsung, Bangalore

Compiler Engineer • Apr 2013 – Apr 2015

	PROBLEM	Samsung's heterogeneous computing platform was evolving, and maintaining the correctness of its OpenCL compiler required broad improvements to the automated random-testing infrastructure across many features. Floating-point expression testing and pointer-expression testing were among the areas needing improvement, alongside other work led by other engineers.
1	SOLUTION	- Developed a floating-point expression testing methodology that validated numerical correctness within a configurable machine-epsilon tolerance. - Designed and implemented a data-flow-analysis-based pointer-range analysis algorithm, integrated it into the random test generator to produce safe, semantically valid pointer-expression test cases, and published an internal technical paper based on this work.
	PROBLEM	Improving the graphics rendering pipeline required identifying and resolving performance bottlenecks in OpenGL shaders.
2	SOLUTION	Collaborated with the graphics rendering team to identify and resolve OpenGL shader performance bottlenecks, improving overall rendering performance.

AMD, Bangalore

Compiler Engineer • Jul 2012 – Mar 2013

	PROBLEM	AMD was migrating its CPU compiler from the Open64 infrastructure to the emerging Clang/LLVM ecosystem, and several critical capabilities were still missing. OpenMP support was a key gap for production adoption, making its design and implementation an important milestone in the transition.
1	SOLUTION	- Proposed an OpenMP support implementation strategy for Clang/LLVM and presented it to the LLVM open-source community for technical review. - Worked with the LLVM community to evaluate design alternatives and converge on an approach aligned with LLVM's architectural direction. - Contributed initial Clang front-end support to recognize OpenMP constructs and generate the corresponding LLVM IR, before the initiative was discontinued due to changing business priorities and responsibility for the upstream implementation transitioned to Intel.

Siemens, Bangalore

Research Engineer • Jul 2008 – Jun 2012

1	PROBLEM	As part of the Siemens Programming Languages Research Group, the goal was to help Siemens business units solve challenges in programming languages, compiler technology, and software development methodology — translating research ideas into engineering solutions that product teams could adopt.
	SOLUTION	• Took ownership of building a new compiler front-end for the Structured Text (ST) language, which is used to program Siemens PLCs. • Joined the product business unit for the duration of the project, took on a technical leadership role for a team with no prior compiler background, and guided them to deliver the product within the required business timeline. • Owned the full engineering lifecycle: requirements specification, architecture and design, prototyping, team mentoring and collaboration, implementation, and delivery.
2	PROBLEM	As heterogeneous computing continues to emerge as a promising computing paradigm, Siemens launched exploratory research into whether application workloads could be automatically analyzed and transformed to run efficiently on heterogeneous platforms such as GPUs.
	SOLUTION	• Collaborated with Siemens research teams across multiple global locations to investigate heterogeneous programming models and source-to-source transformation techniques. • Evaluated multiple architectural approaches through rapid prototyping, including an in-depth study of the CUDA ecosystem and GPU programming methodologies. • Assessed the technical feasibility and trade-offs of the proposed architecture before the research initiative was discontinued due to changing business priorities.
3	PROBLEM	Intel's Many Integrated Core (MIC) architecture was emerging as a potential high-performance computing platform, and Siemens wanted to evaluate its applicability to future business and engineering workloads.
	SOLUTION	• Worked with Intel's programming research team to evaluate the capabilities of the Intel MIC architecture. • Ported representative Siemens workloads to the MIC platform and analyzed their performance characteristics. • Documented the findings and published an internal technical paper summarizing the evaluation.

HP, Bangalore

Compiler Engineer • Sep 2005 – Jun 2008

1	PROBLEM	As the C++ language continues to evolve rapidly, HP's in-house C++ compiler front-end struggled to keep pace due to architectural limitations, making it increasingly difficult to support new C++ features while preserving compiler stability.
	SOLUTION	• Maintained and extended HP's in-house PA-RISC C++ compiler front-end. • Resolved parser issues in a traditional Lex/Yacc-based compiler infrastructure while accommodating modern C++ syntax. • Investigated and resolved semantic issues in C++ templates and other advanced language features.

2	PROBLEM	Maintaining the legacy C++ front-end had become increasingly difficult, so HP began migrating to the industry-standard EDG C++ front-end. The key engineering challenge was preserving HP PA-RISC ABI compatibility while integrating the target-independent EDG front-end.
	SOLUTION	• Collaborated with senior compiler engineers on the EDG integration effort. • Contributed to the design and implementation of an ABI compatibility bridge that lowered EDG-generated ASTs into HP's native ASTs while preserving ABI correctness. • Contributed to AST transformation, code refactoring, bug fixing, and compiler cleanup required for the migration. • Helped successfully integrate the EDG front-end into HP's C++ compiler toolchain.

DACS Software, Bangalore

Software Engineer • Jun 2004 – May 2005

1	PROBLEM	As part of DACS Software's collaboration with CEVA, an existing assembly-to-assembly translation tool needed to be extended to migrate applications from the TEAK DSP architecture to the CEVA DSP architecture. Because CEVA is based on VLIW architecture while TEAK is based on traditional dual-MAC architecture, a straightforward one-to-one instruction mapping could not produce efficient target code.
	SOLUTION	• Contributed to the engineering team extending the assembly-to-assembly translation tool for TEAK-to-CEVA DSP migration. • Implemented an initial instruction-packing strategy that grouped translated instructions into VLIW packets, establishing the baseline for VLIW code generation in the migrated applications. • Investigated and implemented successive optimization techniques to improve the quality and performance of the generated code. • Under tight project timelines, also manually ported selected performance-critical TEAK applications to the CEVA DSP architecture.